TRISHNA mission

Mission characteristics

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2026-10-07

TRISHNA mission overview

Monitoring our ecosystems from space

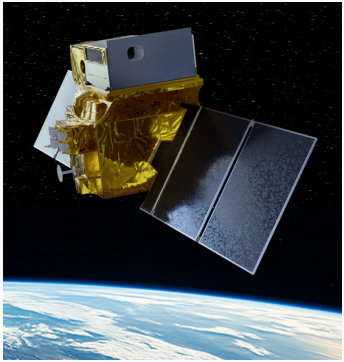


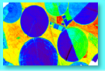
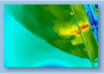
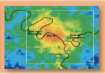
Figure 1: TRISHNA Satellite

- Measuring the **surface temperature** of land surfaces and coastal zones
- Measuring **evapotranspiration**
- Measuring with **high resolution** and **high revisit** frequency.

TRISHNA goals

- Frequent field-scale observations of surface temperatures for continental and coastal areas
- Main Missions : **water in plants** and **temperature of water surfaces**
  - Plants cool down through transpiration, hot vegetation lacks of water
  - Estimation of water consumption from field to watershed scale
  - Decision support tool for irrigation
- Other missions : temperatures in cities, ice and snow, volcanos, hazards and fires

TRISHNA mission themes

Water management Irrigated agriculture Rain-fed agriculture	 	▪ Optimize irrigation ▪ Manage consumption of water ▪ arbitrate water savings	▪ Temperature ▪ Evapotranspiration ▪ Water Stress
Ecosystem health **Drought, fire**	 	▪ follow the droughts and their impact on yield ▪ Diagnosis of vulnerable areas (fire, drought, frost)	▪ Temperature ▪ Evapotranspiration ▪ Water Stress
Coastal waters **Inland waters** **Sea ice**	 	▪ Mixing processes ▪ Water Qualityn algual bloom, discharge of water, pollutants ▪ Ecosystem Productivity (phytoplankton) ▪ Melting and frost Processes	▪ Temperature ▪ Reflectances ▪ Color & indices
Urban heat	 	▪ Urban hydrology ▪ Heat model for buildings ▪ Urban climatology	▪ Temperature ▪ Emissivities ▪ Albedo
Cryosphere Ice sheets, ice caps — Permafrost Snow & glaciers — River & lake ice	 	▪ Snow-melt runoff and debris thickness ▪ Snow cover change & metamorphism, snow energy fluxes ▪ Snow mass, Snow Water Equivalent (SWE), Snow depth	▪ Temperature ▪ Emissivities ▪ Snow & ice albedo ▪ Snow cover
Solid Earth Mineral exploration Volcanology	 	▪ Thermal anomalies ▪ Detection of coal & peat fire ▪ Lava effusion rate	▪ Temperature ▪ Emissivities ▪ Albedo

Indo-French mission overview

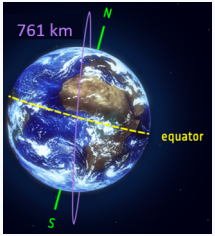
- ISRO/CNES cooperation
- Launch 2028
- 5-year lifetime
- Design drivers: ecosystem stress and water use; coastal & inlands waters



TRISHNA mission characteristics

Orbit overview

- Altitude: 761 km, Sun synchronous
- Swath: +/-38° $\longrightarrow$ 1030 km
- Local time: 12:30 +/-5 min descending node
- Orbit cycle:
  - 8 days geometric revisit time,
  - 3 days revisit with different viewing angles



TRISHNA resolution

- **60m** ground resolution

Visible to infrared electromagnetic spectrum

- 0.4 µm to 0.75 µm: **Visible**
- 0.75 µm to 1.3 µm: **Near Infrared** (NIR)
- 1.3 µm to 3 µm: **Short Wave Infrared** (SWIR)
- 3 µm to 8 µm: **Middle infrared** (MIR)
- 8 µm to 14 µm: **Thermal Infrared** (TIR)
- 14 µm to 100 µm: **Far Infrared** (FIR)

TRISHNA spectral bands

- **VNIR-SWIR 7 bands**: day-time acquisition
- **TIR 4 bands**: day-time/night-time acquisition



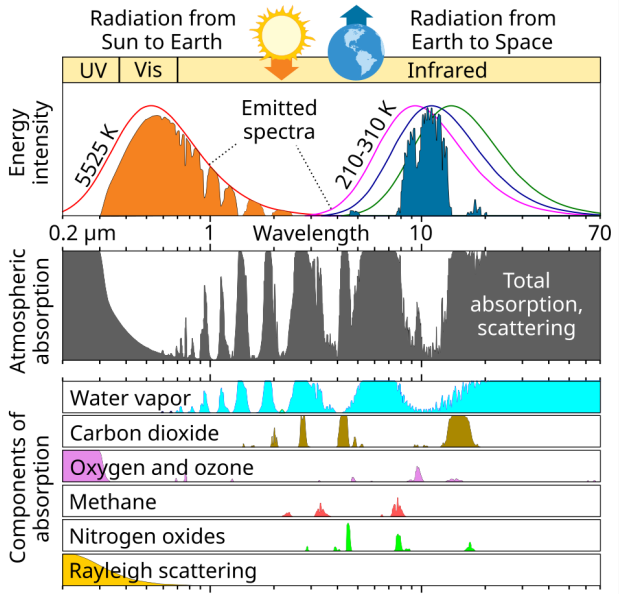
TRISHNA VNIR-SWIR spectral bands

Band name	λ (µm)	Band width (nm)	Purpose
`Blue`	0.485	70	Detection of low clouds
`Green`	0.555	70	Coastal, sediments, snow
`Red`	0.670	60	Vegetation (LAI, fCOVER, NDVI,...)
`NIR`	0.860	40	Vegetation (LAI, fCOVER, NDVI,...)
WV	0.910	30	Water vapour content estimation
Cirrus	1.380	30	Detection of thin cirrus clouds
`SWIR`	1.610	100	AOD, snow/cloud discrimination, vegetation stress , burnt areas

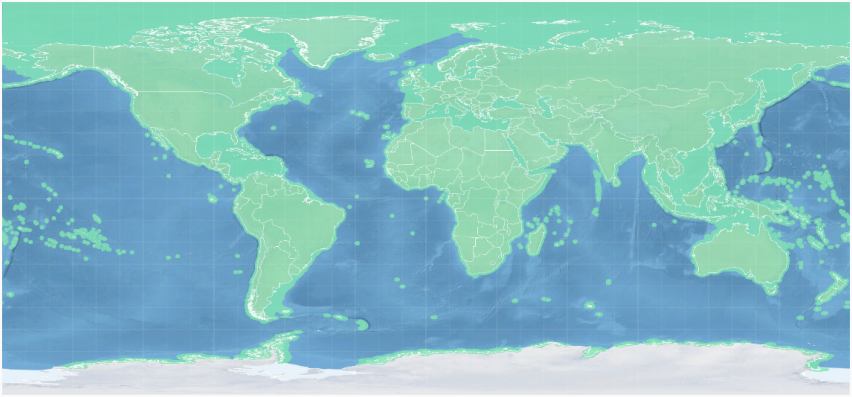
TRISHNA TIR spectral bands

Band name	λ (µm)	Band width (nm)	Purpose
`TIR 1`	8.65	350	Temperature/emissivity separation
`TIR 2`	9.0	350	Temperature/emissivity separation
`TIR 3`	10.6	700	Split-window
`TIR 4`	11.6	1000	Split-window

Atmospheric transmission

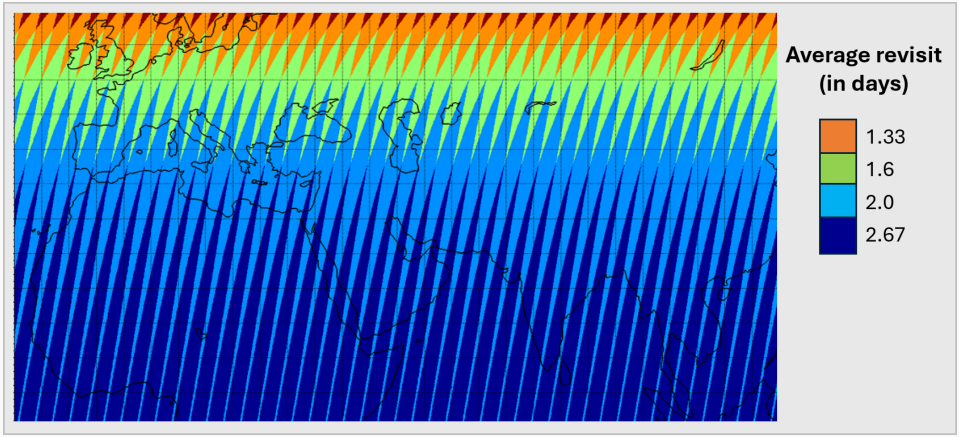


TRISHNA global coverage land + coastal



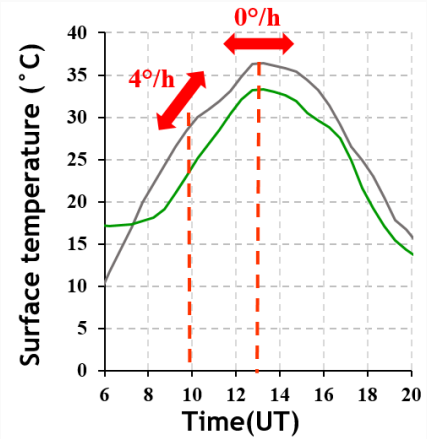
- Continental surfaces
- Coastal areas 100 km from the coastline
- Coastal waters with bathymetry < 250m + Closed seas + Mediterranean Sea
- Antarctica coastline

TRISHNA revisit



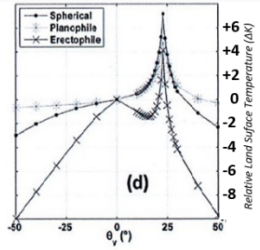
TRISHNA overpass time

- Overpass time: 12:30 PM & AM at the Equator
- Minimizing the sensitivity of LST measurement to overpass time

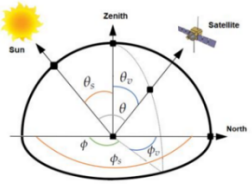


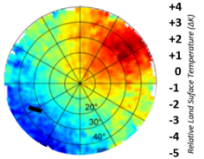
TRISHNA observation angles

- Different observation angles, up to 38 deg



SCOPE Model (Duffour et al., 2015)





Measurements over Maritime pines

TRISHNA Launch

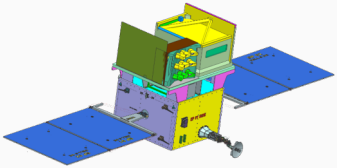
- Launch Vehicle : PSLV
- Launch from SDSC-SHAR Sriharikota, Andhra Pradesh, India
- Launch date: 2028

PSLV Launcher

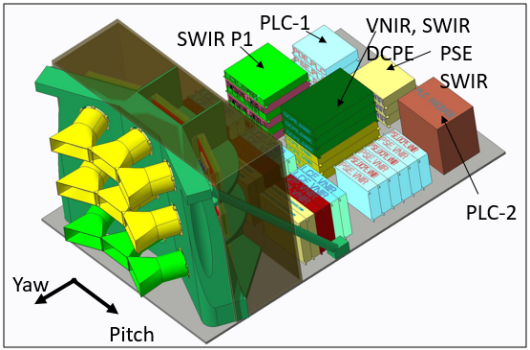
Launch Site

TRISHNA satellite characteristics

- I-1k bus (ISRO flight heritage)
- Mass: 700 kg (S/C) + 330 (P/L)
- Power generation: 2kW
- Pointing accuracy: +/- 0,05°
- Pointing stability: 5.0 x 10-04 °/s
- TM/TC: S-band, 64 / 4kbps
- X-band data rate: 960 Mbps
- Mass memory: 1.4 Tb
- Mission lifetime: 5 years
- Consumables for: 7 years

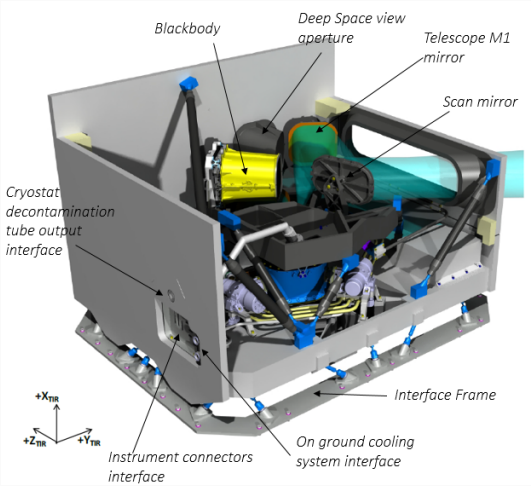


TRISHNA VIS – NIR – SWIR instrument



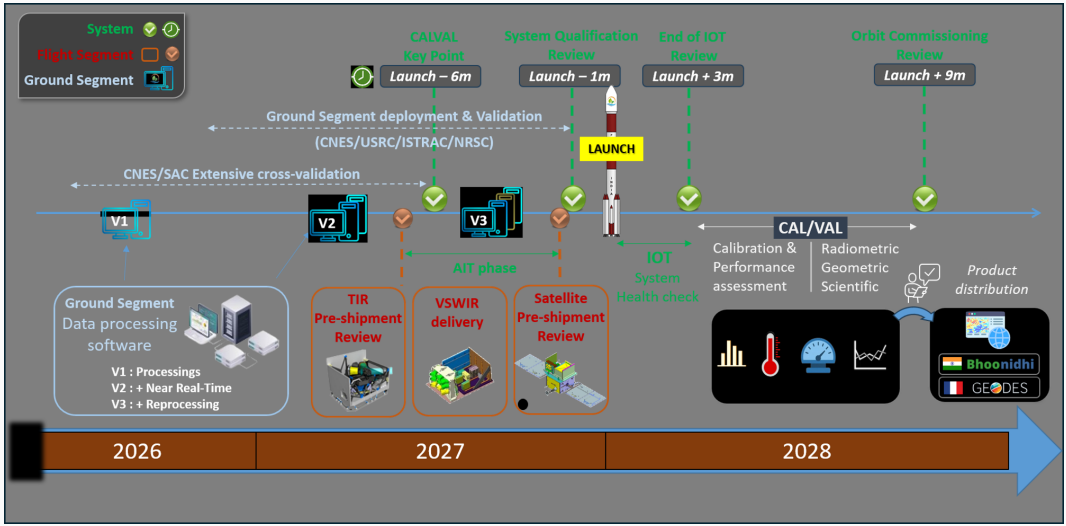
- Developed by ISRO / SAC
- Optical Heads 9
- Acquisition Pushbroom
- Mass 207 kg
- Power 300W/350W
- VIS & NIR bands 5
- SWIR bands 2
- Delivery 2027

TRISHNA TIR instrument

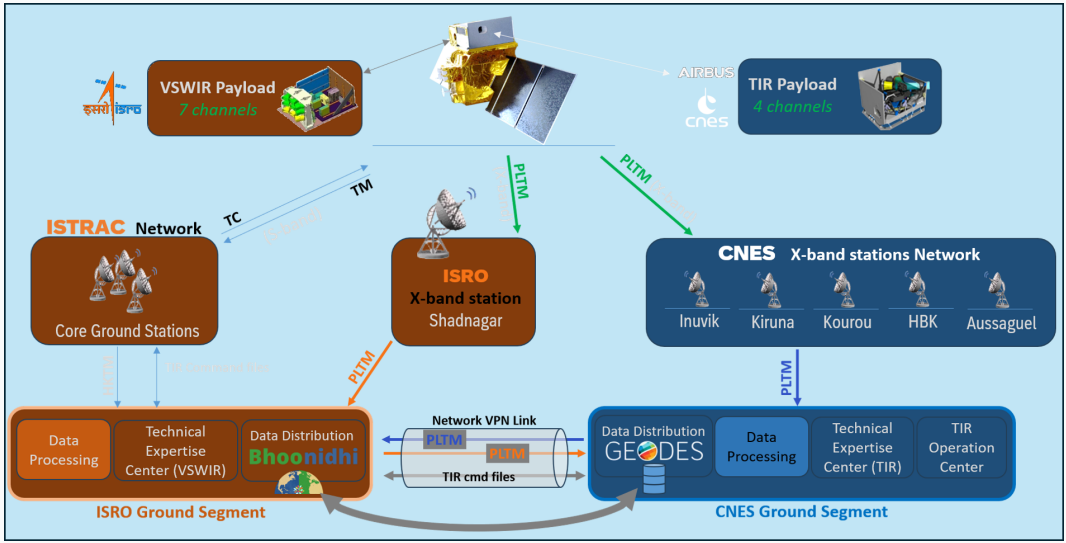


- Developed by AIRBUS DEFENCE AND SPACE
- Acquisition Across track scanner
- Mass 210 kg
- Power 300W/350W
- Spectral bands 4 TIR
- Wavelength Center 8.65 / 9.0 /10.6 / 11.6µm
- Delivery 2027

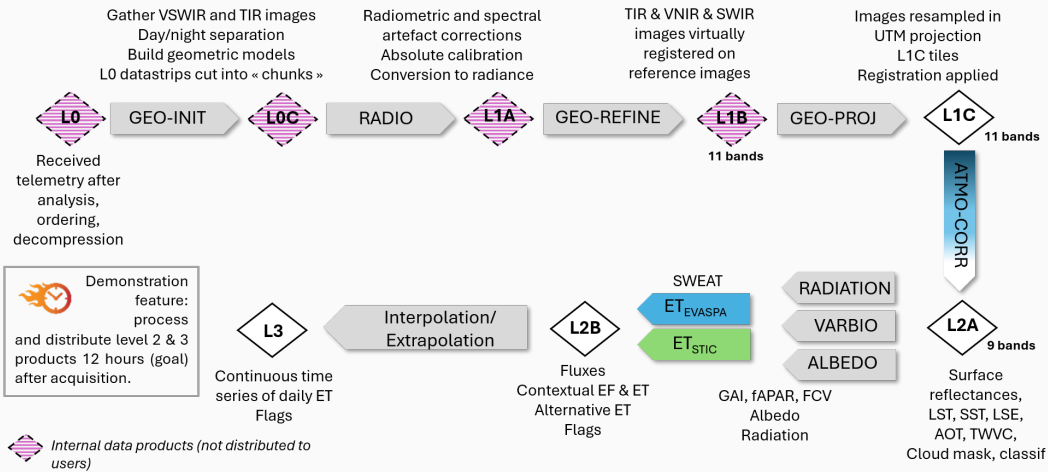
TRISHNA Milestones



Indo-French ground segment



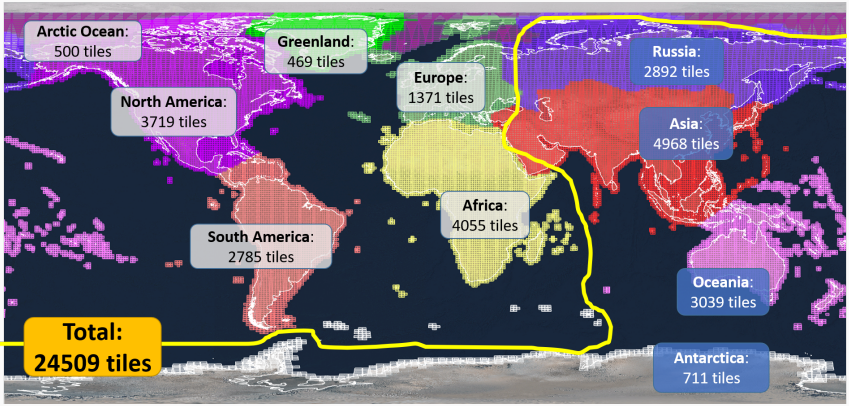
TRISHNA processing chain



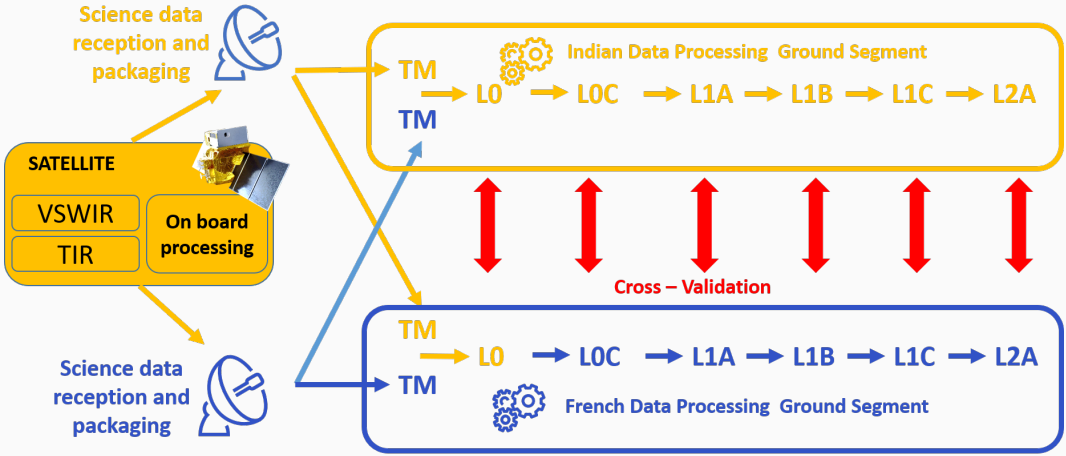
TRISHNA Production and data distribution

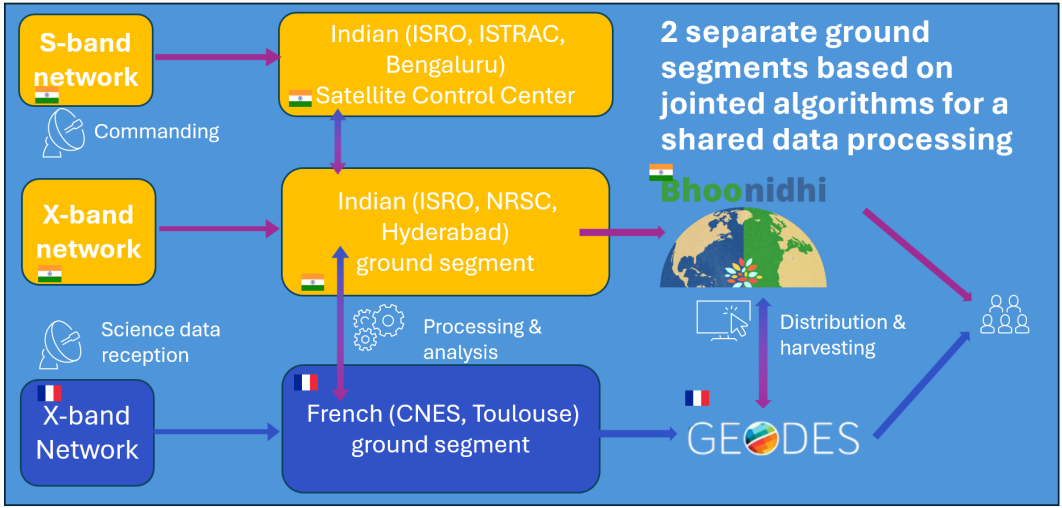
- Free & open data policy
- Objective:
  - Products available 12h after acquisition for L2A & L2B
  - Products available 15h after acquisition for L3
- Distributed products per day: 81623
- Significant volume: ~10 Po over 7 years
- Indo-french work share: Production, Cataloguing
- Cross-validation

TRISHNA tile repartition



TRISHNA cross validation




S-band network
Commanding
Indian (ISRO, ISTRAC, Bengaluru) Satellite Control Center
2 separate ground segments based on jointed algorithms for a shared data processing
X-band network
Indian (ISRO, NRSC, Hyderabad) ground segment
Bhoonidhi
Science data reception
Processing & analysis
Distribution & harvesting
X-band Network
French (CNES, Toulouse) ground segment
GEODES

TRISHNA products

- The **radiance** is the radiant flux in a beam per unit area and solid angle of that beam, and is expressed in the SI units $W.m^{-2}.sr^{-1}$.
- The **reflectance** is the ratio of the radiant exitance ($W.m^{-2}$) with the irradiance ($W.m^{-2}$).

$$\rho_\lambda = \frac{\text{Radiant exitance}}{\text{Irradiance}} \in [0, 1]$$

Distributed products overview

- **Level 1**: Top-Of-Atmosphere (TOA) radiances and brightness temperatures
- **Level 2**: Bottom-Of-Atmosphere (BOA) reflectances and products

Level 1C products

- TOA radiances x7 VNIR/SWIR bands
- TOA radiances x4 TIR bands $\Rightarrow$ Brightness temperatures
- $\Rightarrow$ *Radiometrically and geometrically calibrated*
- $\Rightarrow$ *Orthorectified and resampled on a uniform spatial grid (Sentinel-2 tiles, Copernicus DEM)*

What is brightness temperature?

What is a black body ?

- A **black body** is an idealized physical body that absorbs all incident electromagnetic radiation, regardless of frequency or angle of incidence.

What is brightness temperature?

It is the temperature of a black body that would have the same radiance as the radiance actually observed with the radiometer.

Spectral radiance of a black body

Defined by Planck's law

$$B_\lambda(T) = \frac{2hc^2}{\lambda^5} \frac{1}{\exp \frac{hc}{\lambda \sigma_B T} - 1}$$

where

- h Planck's constant
- c light velocity
- σ_B Boltzmann's constant
- T temperature
- λ wavelength

From Level 1C to level 2 products: How to define TOA radiance?

Radiative transfer equation in the atmosphere:

$$L_{\lambda,TOA} = L_{\lambda}^{atm\uparrow} + \left[\rho_{\lambda} \times L_{\lambda}^{atm\downarrow} + \epsilon_{\lambda} \times B(T)\right] \times \tau_{\lambda}$$

- ρ_{λ} reflectance
- ϵ_{λ} land surface emissivity
- T land surface temperature
- $L_{\lambda}^{\downarrow}$ downward atmospheric radiance
- $L_{\lambda}^{\uparrow}$ upward atmospheric radiance
- τ_{λ} atmospheric transmittance
- B_{λ} spectral radiance of a black body (Planck's law)

Focus on emissivity

- For isothermal, homogeneous emitters, the emissivity can be defined as the ratio of the actual emitted radiance to the radiance emitted from a black body at the same thermodynamic temperature.
- For heterogeneous surfaces that are not in thermal equilibrium, the concept of directional emissivity is ambiguous.

See [@norman_terminology_1995]

Compute BOA radiance

$$L_{\lambda}^{BOA}(\theta,\phi) = \frac{L_{\lambda}^{TOA}(\theta,\phi) - L_{\lambda}^{atm\uparrow}(\theta,\phi)}{\tau_{\lambda}(\theta,\phi)}$$

where

- θ zenith angle and ϕ azimuth angle
- L_{λ}^{TOA} TOA radiance
- $L_{\lambda}^{atm\uparrow}$ upwelling atmospheric radiance
- τ_{λ} atmospheric transmittance

Top-Of-Canopy Radiance

The TOC radiance includes the surface emission term and the surface reflected downward atmospheric emission term

$$L_{\lambda}^{BOA}(\theta,\phi) = \left[\epsilon_{\lambda}(\theta,\phi)B_{\lambda}(T(\theta,\phi)) + \rho_{\lambda}(\theta,\phi)L_{\lambda}^{atm\downarrow}(\theta,\phi)\right]$$

where

- θ zenith angle and ϕ azimuth angle
- ϵ_{λ} land surface emissivity
- ρ_{λ} surface reflectance
- T land surface temperature
- $L_{\lambda}^{atm\downarrow}$ downward atmospheric radiance
- B_{λ} Planck's law

Remarks

- $L_{\lambda}^{atm\downarrow}$, $L_{atm\lambda}^{\uparrow}$ and τ_{λ} are estimated with an atmospheric radiative transfer model using atmospheric profiles of pressure, air temperature and relative humidity as inputs.
- The land surface temperature, emissivity and reflectance depend on (θ, ϕ)
- Emissivity is an intrinsic property of the surface.
- The surface temperature varies with irradiance and local atmospheric conditions.

Level 2A products

After atmospheric correction:

- Land & Water Surface reflectances x5 VNIR/SWIR bands
- Land Surface Temperature and Sea Surface Temperature
- Land Surface Emissivity x4 TIR bands
- Cloud mask, Total Water Vapor Content
- *Precision NeDt 0.2K at instrument output, AKA 0.5K*

Level 2B products: Biophysical variables

- Surface albedo
- Vegetation variables
- Instantaneous radiation
- Daily evapotranspiration

Computed for clear pixels

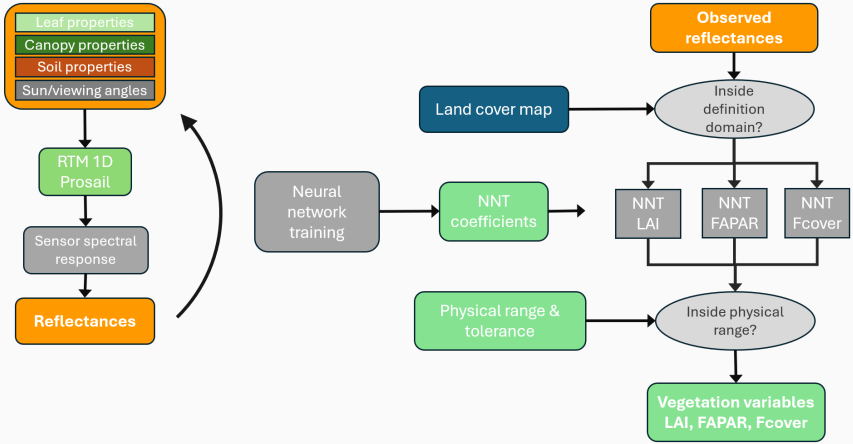
Level 2B: Broadband surface albedo

- Compute blue-sky albedo which combines black-sky albedo (direct) and white-sky albedo (diffuse)
- Provide two boradband albedo products:
  - **Instant albedo**
  - **“Climate” albedo** using from a climatology or from the history file

Level 2B: Vegetation variables

- **LAI**: Leaf Area Index
- **FAPAR**: Fraction of photosynthetically active radiation absorbed by the canopy
- **Fcover**: Fraction cover in a pixel

Level 2B: BVnet algorithm



See [@weiss_gai_2024]

Level 2B: Instantaneous radiation

- Shortwave downwelling radiation
- Diffuse fraction
- Longwave downwelling radiation

Level 2B: Instantaneous shortwave radiation

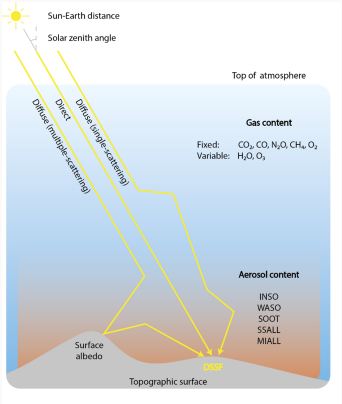


Figure 2: TRISHNA Satellite

$$R_S^{\downarrow} = E_{S,direct}^{\downarrow} + E_{S,diffuse}^{\downarrow}$$

$$R_S^{\downarrow} = E_{S,direct}^{\downarrow} + (1 - f) \times E_{S,direct}^{\downarrow}$$

$$R_S^{\downarrow} = E_{TOA} T_{\text{direct}} + E_{TOA}(T_{SS} + T_{MS})$$

Where:

- E radiation
- T transmittance

$$R_L^{\downarrow} = \sigma_B \epsilon_{sky} T_{sky}$$

Where:

- σ_B Boltzmann's constant
- ϵ_{sky} Sky emissivity estimated from air temperature and dewpoint temperature
- T_{sky} Sky temperature estimated from air temperature and dewpoint temperature

Level 2B: Daily evapotranspiration

- Contextual daily evapotranspiration (EVASPA)
    - Instantaneous evaporative fraction
    - Daily evapotranspiration
    - Water stress indicator
- Single-pixel daily evapotranspiration (STIC)
    - Instantaneous evaporative fraction
    - Daily evapotranspiration
    - Water stress indicator

See more details in the rest of the tutorial.

Level 3 products

- Continuous time series of daily evapotranspiration: Each day L3 product is provided even for day without TRISHNA acquisition
- Computed using interpolation and extrapolation using auxiliary data

See more details in the rest of the tutorial.

International cooperation

Collaboration in Europe

Organismes de tutelle

www.cnrs.fr
www.meteofrance.fr
www.cnes.fr
www.ofb.gouv.fr
www.ifremer.fr
www.inrae.fr
www.ird.fr
www.cerema.fr



Universités de tutelle



Collaboration in India



Earth Observation missions in synergy

